Practical 8 – Porosity determination

Introduction

Porosity is a fundamental property of rocks and sediments. It measures the capacity of a granular solid to store fluid (gas or liquid). Hence it determines the capacity of a reservoir (petroleum or water aquifer).

Porosity is defined as (vol. voids) / (vol. voids + vol. solids), expressed as a percentage or a fraction.

Learning outcomes

By the end of this practical, you should be able to:

- explain methods to estimate total porosity
- describe a simple analytical technique, write up results, and interpret them scientifically

Materials

In a group of 3 or 4, you are provided with:

- two rock cores, a sandstone (red) and a limestone (white)
- loose, 100% quartz sand in a plastic bottle
- plastic beaker
- calipers
- weighing scale

Task

1. To determine the porosity of the core plugs:

- Calculate the bulk volume
- Weigh the plug
- Calculate the bulk density
- Using the grain density provided, calculate the grain volume
- Calculate the pore volume
- Calculate porosity

2. To determine the porosity of the loose sand:

- Weigh the container
- Fill the container with loose sand, weigh and subtract the weight of the container
- Calculate the bulk volume
- Calculate the bulk density
- Using the grain density provided, calculate the grain volume
- Calculate the pore volume
- Calculate porosity

Record and compare your results in the table below:

Sample	Weight	Bulk Volume	Grain volume	Grain density	Pore volume	Porosity

Equations

volume of cylinder (SI cm): $V = \pi r^2 \times h$

density (SI g/cm^3): $\rho = M/V$

grain volume (SI %): $V_{pore} = V_{bulk} - V_{grain}$

porosity (SI %): V_{pore}/V_{bulk}